HOMEWORK SET 3

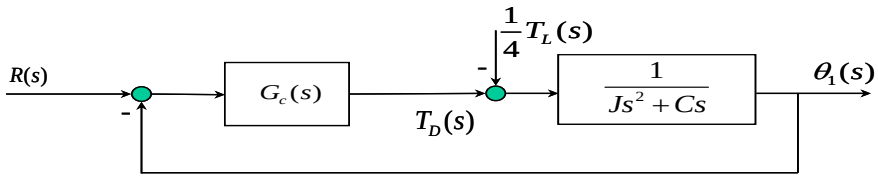
Modeling, Identification and Control of Mechatronic Systems **Fall 2022**

OUT: 17/10/22 DUE: 24/10/22

Use separate sheets to answer each problem with clear problem number, figures, equations and illustrations. **Please submit the answer sheets together with Matlab codes and Simulink model (Matlab2015a Version or below)** used in your problem analysis when you hand in your homework.

PROBLEM #1(40%)

Consider the control of ECP emulator with rigid belts shown below



where J=0.00041 and C=0.00076. You are required to design a **controller** $G_c(s)$ to meet the following transient performance specifications

(P1). The resulting CL system should have an OS% less than 16.3%.

(P2). 2% settling time t_s should be smaller than 0.8 second.

To solve the problem, you are asked to follow the following procedure:

(A) Verify that the poles -8±8j and -40 are in the performance region, and determine a **lead controller,** ($C_{lead}(s) = K_c \frac{s+\beta_1}{s+\alpha_1}$ in which K_c, α_1, β_1 are controller parameters) such that the above transient performance requirements are satisfied.

(B) Plot the root locus of your design via MATLAB. Obtain all closed-loop poles of your design and show them on the root locus plot.

(C) Obtain the unit step response of your design via MATLAB. Verify if your design achieves the goal of having an OS% less than 16.3% and 2% settling time t_s less than 0.8 second. If not, explain the reasons.

In addition to meet the above transient performance requirements, the controller $G_c(s)$ is also required to meet the following steady-state performance requirement

(P3). The steady-state tracking error for a unit constant disturbance input torque T_L should be less than 0.05.

To solve the problem, you are asked to follow the following procedure:

(D) Obtain the steady-state tracking error of the lead compensator designed in the above for a unit constant disturbance input torque T_L. Check if the lead compensator meets the steady-state performance requirement P3. Verify your answer using the response of the closed-loop system under a unit disturbance input torque via MATLAB.

(E) If P3 is not satisfied, add a lag compensator ($C_{lag}(s) = \frac{s+n\alpha_2}{s+\alpha_2}$ in which n, α_2 are controller parameters) so that the resulting **lead-lag compensator** meets all the requirements.

(F) Obtain the closed-loop transfer functions from the reference input $R(s)$ and the disturbance input T_L to the output $\theta_1(s)$ respectively. Locate the zeros and poles of the two closed-loop transfer functions on

the complex plan respectively. Plot the root locus of your design via MATLAB and show all closed-loop poles on the root locus plot.

(G) Obtain the **unit step response** of your design via MATLAB. Verify if your design achieves the goal of having an OS% less than 16.3% and 2% settling time t_s less than 0.8 second. If not, explain the reasons. In addition, obtain the response of the closed-loop system under a **unit disturbance input torque** via Matlab and verify if the steady-state performance requirement P3 is satisfied.

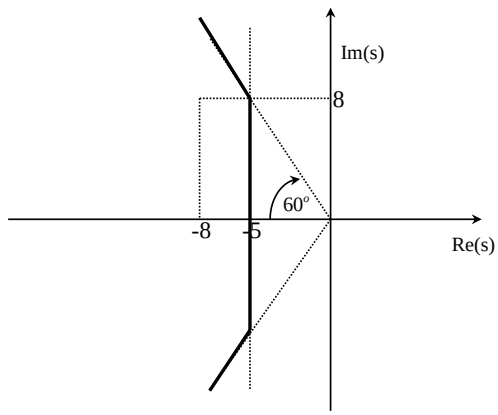
Solutions:

(A). To meet transient performance specifications, we requires:

$$\text{OS\%} < 16.3\% \;\Rightarrow \zeta > \sqrt{\frac{\ln^2(0.163)}{\pi^2+\ln^2(0.163)}} = 0.5, \quad \varphi < \tan^{-1}\left(-\frac{\pi}{\ln(0.163)}\right) = 60^\circ$$

$$t_s \le 0.8\,\text{sec} \;\Rightarrow\; \alpha > \frac{4}{t_s} = 5 \tag{3}$$

As shown below, -8 ± 8*j* and –40 are in the performance region:



With a lead compensator $C_{lead}(s) = K_c\dfrac{s+\beta_1}{s+\alpha_1}$, the resulting CLTFs are:

$$\theta_1(s) = \frac{P\,C_{lead}}{1+P\,C_{lead}}R(s) + \frac{-\frac{1}{4}P}{1+P\,C_{lead}}T_L(s)$$

$$= \frac{\dfrac{1}{(Js^2+Cs)}\cdot\dfrac{K_c(s+\beta_1)}{(s+\alpha_1)}}{1+\dfrac{1}{(Js^2+Cs)}\cdot\dfrac{K_c(s+\beta_1)}{(s+\alpha_1)}}R(s) + \frac{-\frac{1}{4}\dfrac{1}{Js^2+Cs}}{..}T_L(s)$$

$$= \underbrace{\frac{K_c(s+\beta_1)}{Js^3+(J\alpha_1+C)s^2+(C\alpha_1+K_c)s+K_c\beta_1}}_{G_{CLR}(s)}R(s) + \underbrace{\frac{-\frac{1}{4}(s+\alpha_1)}{Js^3+(J\alpha_1+C)s^2+(C\alpha_1+K_c)s+K_c\beta_1}}_{G_{CLT}(s)}T_L(s) \tag{1}$$

From (1), the actual CLCE is:

$$s^3 + \frac{J\alpha_1+C}{J}s^2 + \frac{C\alpha_1+K_c}{J}s + \frac{K_c\beta_1}{J} = 0 \tag{2}$$

As seen from (2), since we have three parameters, K, β_1, α_1 to choose, we can arbitrarily place three CL poles. So choose $p_{CL1,2d} = -8 \pm 8j$ and $p_{CL3d} = -40$ as required.

Desired CLCE:

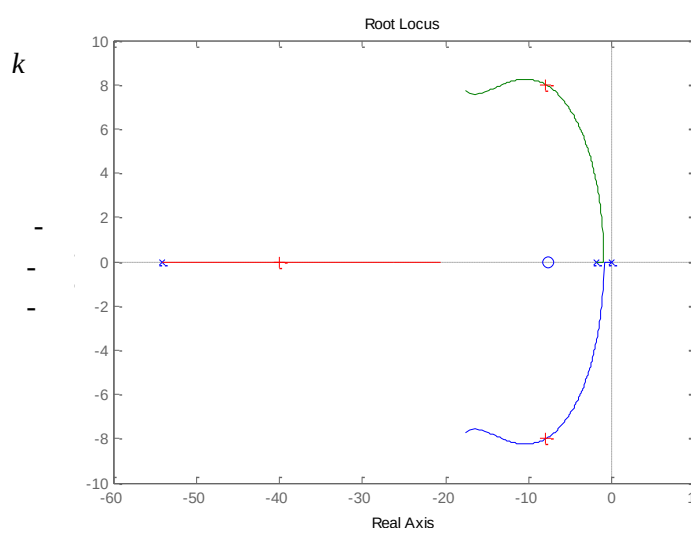
$$\begin{aligned}(s - p_{CL1d})(s - p_{CL2d})(s - p_{CL3d}) &= (s+8-8j)(s+8+8j)(s+40) \\ &= s^3 + 56s^2 + 768s + 5120\end{aligned} \tag{3}$$

Comparing (2) with (3):

$$\begin{aligned}\frac{J\alpha_1 + C}{J} = 56 \quad &\Rightarrow \alpha_1 = 54.15 \\ \frac{C\alpha_1 + K_c}{J} = 768 \quad &\Rightarrow K_c = 0.274 \\ \frac{K_c\beta_1}{J} = 5120 \quad &\Rightarrow \beta_1 = 7.67\end{aligned} \tag{4}$$

(B). With the following MATLAB file, one can obtain the root locus plot of the design as shown in Fig.1, in which the CL poles are indeed placed at the three desired location.

```
clf; clear
J=0.00041; C=0.00076; numP=1; denP=[J,C,0];%plant
w1=-8+8*j; w2=-8-8*j; w3=-40; %desired CL pole locations
denCLd=poly([w1,w2,w3]); %desired CLCE
alpha1=denCLd(2)-denP(2)/denP(1);
Kc=(denCLd(3)*denP(1)-alpha1*denP(2));
beta1=denCLd(4)*denP(1)/Kc;
numC=Kc*[1,beta1]; denC=[1,alpha1]; %The lead controller C(s)
numTd=-0.25*[1, alpha1]; %Numerator of Disturbance CLTF
[numL,denL]=series(numC,denC,numP,denP); %Form L(s)=C(s)P(s), open-loop TF
[numT,denT]=cloop(numL,denL,-1); %Find the closed-loop TF T(s)
step(numT,denT); %plot unit step response of lead controller
pause
K=0:0.01:1.5;
rlocus(numL,denL,K);
[k,poles]=rlocfind(numL,denL)
pause
step(numTd,denT); %plot unit step disturbance response of lead controller
```



k =
0.9994

poles =
40.0114
7.9943 + 7.9988i
7.9943 - 7.9988i

Fig.1

(C) As shown in Fig.2, the closed-loop system has an OS% more than 20% but a 2% settling time less than 0.8 second. The design thus satisfies the settling time requirement but not the OS% due to the added zero at -7.67 from the lead controller.

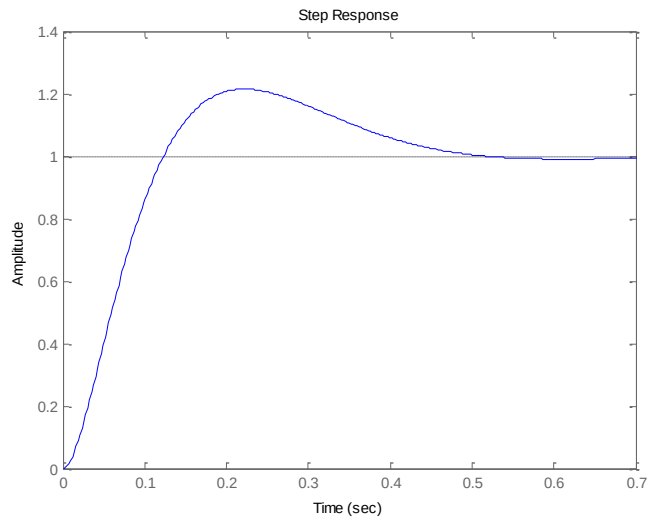


Fig.2

(D). With the lead compensator (4), the steady-state error for a unit step disturbance torque of $T_L(s) = \frac{1}{s}$ (or $T_L(t) = 1$) is:

$$e_{ss} = \lim_{s \to 0} s\, G_{CLT}(s)\, T_L(s) = G_{CLT}(0) = \frac{-\frac{1}{4}\alpha_1}{K_c\, \beta_1} = -6.45 \qquad (5)$$

which is more than the desired level of $|e_{ssd}(t)| \le 0.05$. Such an assertion is also confirmed by the response of the closed-loop system under a unit disturbance input torque via Matlab shown in Fig.3. Thus a lag compensator (or other controller structure) may be needed to boost the static gain to reduce e_{ss} .

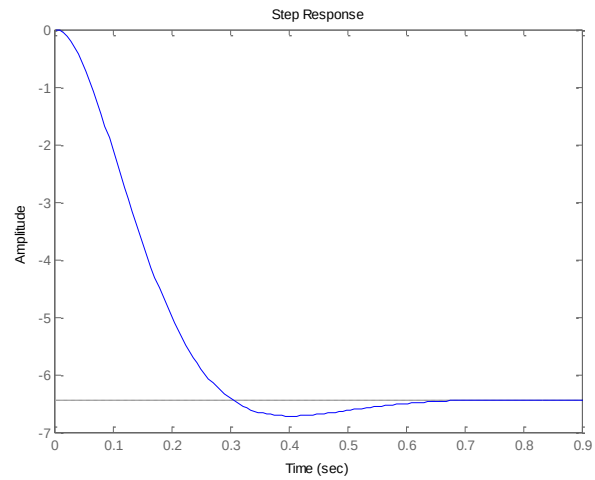


Fig.3

(E). With the addition of a lag compensator, the controller becomes:

$$C(s) = C_{lead}(s)\, C_{lag}(s) = K_c \frac{s+\beta_1}{s+\alpha_1} \cdot \frac{s+n\alpha_2}{s+\alpha_2} \tag{6}$$

With this controller,

$$G_{CLT}(s) = \frac{-\frac{1}{4}P}{1+P\,C} = \frac{-\frac{1}{4}(s+\alpha_1)\,(s+\alpha_2)}{(Js^2+Cs)\,(s+\alpha_1)\,(s+\alpha_2)+K_c(s+\beta_1)\,(s+n\alpha_2)} \tag{7}$$

$$\Rightarrow G_{CLT}(0) = \frac{-\frac{1}{4}\alpha_1\,\alpha_2}{K_c\beta_1\,n\,\alpha_2} = \frac{-\frac{1}{4}\alpha_1}{K_c\beta_1} \cdot \frac{1}{n} = -\frac{6.45}{n} \tag{8}$$

To have $|e_{ssd}| \le 0.05$ for a unit step disturbance, we need $|G_{CLT}(0)| \le 0.05$

$$\therefore \qquad n \ge \frac{6.45}{0.05} = 129 \qquad \text{let} \qquad \text{n = 129} \tag{9}$$

Pick the zero of the lag compensator $z_2 = -n\alpha_2$ to be 20 times slower than the dominant poles:

$$n\alpha_2 = \frac{1}{20} \times 8 = 0.4 \quad \Rightarrow \quad \alpha_2 = \frac{0.4}{n} = 0.0031 \tag{10}$$

$$\therefore \quad C(s) = \frac{0.274(s+7.67)}{s+54.15} \cdot \frac{s+0.4}{s+0.0031} = \frac{0.274\,s^2+2.209\,s+0.84}{s^2+54.15s+0.168} \tag{11}$$

(F). With the lead-lag compensator (11), the resulting CLTFs are:

$$\begin{aligned}\theta_1(s) = \frac{P\,C}{1+P\,C}R(s) + \frac{-\frac{1}{4}P}{1+P\,C}T_L(s) &= \underbrace{\frac{K_c(s+\beta_1)(s+n\alpha_2)}{(Js+C)s(s+\alpha_1)(s+\alpha_2)+K_c(s+\beta_1)(s+n\alpha_2)}}_{G_{CLR}(s)} R(s) \\ &+ \underbrace{\frac{-\frac{1}{4}(s+\alpha_1)(s+\alpha_2)}{(Js+C)s(s+\alpha_1)(s+\alpha_2)+K_c(s+\beta_1)(s+n\alpha_2)}}_{G_{CLT}(s)} T_L(s)\end{aligned} \tag{12}$$

The 4 CL poles are:

$$p_{CL1} = -40.01, \quad p_{CL2,3} = -7.79 \pm 8.22j, \quad p_{CL4} = -0.403$$

The 2 zeros of $G_{CLR}(s)$ are:

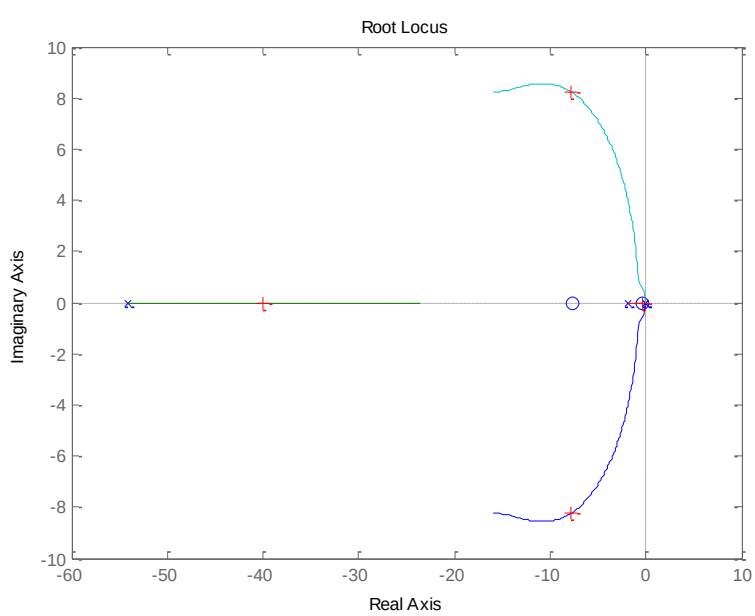
$$z_1 = -7.67, \quad z_2 = -0.4$$

The 2 zeros of $G_{CLT}(s)$ are:

$$-54.15, \quad -0.0031$$

The root locus is shown in Fig.4

```
clf; clear
J=0.00041; C=0.00076; numP=1; denP=[J,C,0];%plant
w1=-8+8*j; w2=-8-8*j; w3=-40; %desired CL pole locations
denCLd=poly([w1,w2,w3]); %desired CLCE
alpha1=denCLd(2)-denP(2)/denP(1);
Kc=(denCLd(3)*denP(1)-alpha1*denP(2));
beta1=denCLd(4)*denP(1)/Kc;
GCLTd=0.05; n=alpha1/(4*Kc*beta1)/GCLTd;
alpha2=abs(real(w1))/(20*n);
numC=Kc*conv([1,beta1],[1,n*alpha2]);
denC=conv([1,alpha1],[1,alpha2]); %The lead-lag controller C(s)
numTd=-0.25*conv([1, alpha1],[1,alpha2]); %Numerator of Disturbance CLTF
[numL,denL]=series(numC,denC,numP,denP); %Form L(s)=C(s)P(s), open-loop TF
[numT,denT]=cloop(numL,denL,-1); %Find the closed-loop TF T(s)
step(numT,denT); %plot unit step response of lead controller
pause
K=0:0.01:1.5;
rlocus(numL,denL,K);
[k,poles]=rlocfind(numL,denL)
pause
step(numTd,denT); %plot unit step disturbance response of lead controller
```



$k =$ 1.0094

poles =
-40.0118
-7.7944 + 8.2215i
-7.7944 - 8.2215i
-0.4026

Fig.4

(G) As shown in Fig.5, the closed-loop system has a 2% settling time less 0.8 second as expected but has an OS% more than 20% due to the zero at $z_1 = -7.67$ added by the lead-lag controller. Furthermore, although the steady-state error caused by the unit disturbance input torque shown in Fig.6 is indeed within 0.05, the transient response takes much longer time (more than 10 seconds) than that in Fig.5 due to the slow closed-loop pole at $p_{CL4} = -0.403$ added by the lag compensation.

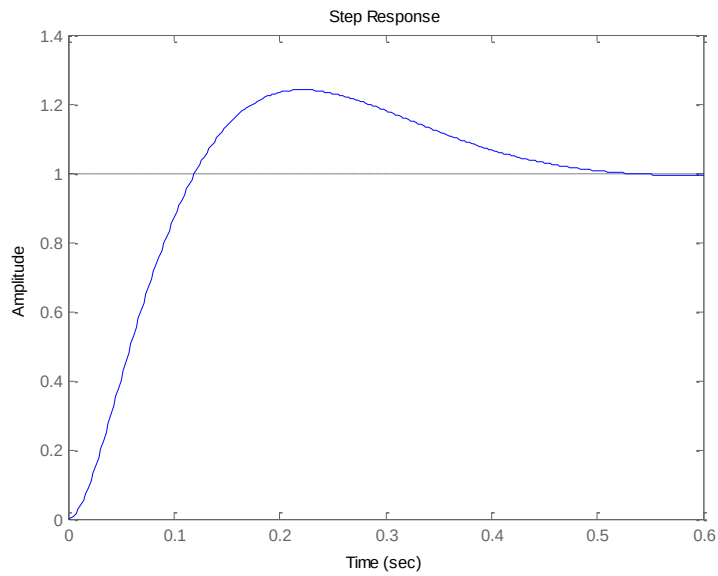


Fig.5

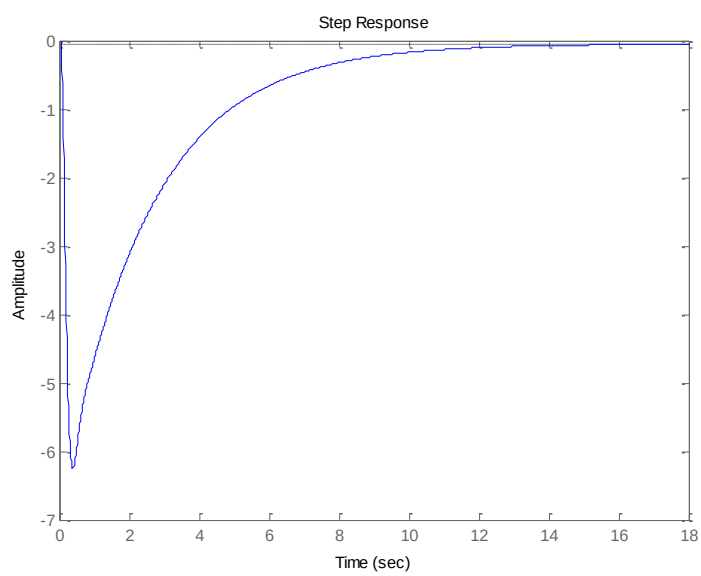


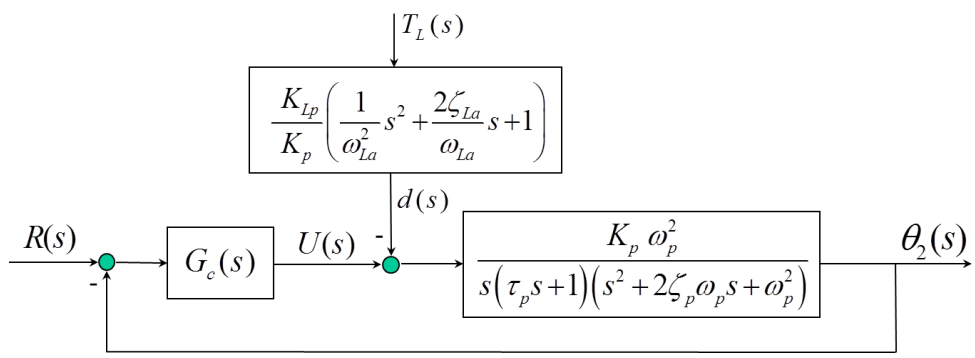
Fig.6

PROBLEM #2 (35%)

When the lead controller designed in Problem 1 is implemented on the actual system, it is found that significant oscillations exist, which cannot be explained based on the analysis in Problem 1. Further analysis reveals that those oscillations are due to the neglected flexible modes of the driving belts, which is shown below. Specifically, when the flexibility of the driving belts is modeled, the resulting plant transfer functions would be of order 4 with a pair of lightly damped poles added. In fact, with the input voltage to the motor being the control input to the physical plant and the output the rotational angle of the load disk θ_2, it is shown that the physical plant can be modeled by

$$\theta_2(s) = \underbrace{\frac{K_p}{s\left(\tau_p s+1\right)\left(\frac{1}{\omega_p^2}s^2+\frac{2\zeta_p}{\omega_p}s+1\right)}}_{G_{\theta U}(s)} U(s) - \underbrace{\frac{K_{Lp}\left(\frac{1}{\omega_{La}^2}s^2+\frac{2\zeta_{La}}{\omega_{La}}s+1\right)}{s\left(\tau_p s+1\right)\left(\frac{1}{\omega_p^2}s^2+\frac{2\zeta_p}{\omega_p}s+1\right)}}_{G_{\theta T}(s)} T_L(s)$$

where the flexible mode of the driving belts is modeled by the resonance frequency ω_p with a very small damping ratio ζ_p. Thus, a more realistic description of the control of ECP emulator should be



(1)

In the following, assume K_p=1316, τ_p=0.54, ω_p =80, and ζ_p=0.02 .

(A) Consider the lead controller designed in Problem 1 given by

$$G_{Clead}(s) = \frac{0.274\left(s+7.67\right)}{s+54.15} \tag{2}$$

which places the three closed-loop poles at -8±8j and -40 should the flexible modes of the ECP emulator be neglected. However, in reality, due to the existence of the flexible mode, even with the natural frequency of the flexible mode ω_p =80 is quite higher than the intended dominant CL poles at-8±8j, the actual CL system (i.e., the system in (1)) with this lead controller becomes unstable; *this phenomenon is sometimes referred to as the "spill-over effect" since it seems that some of the "control energy" has spilled over into the lightly damped flexible mode and caused it to be excited*. Verify this instability by using MATLAB to obtain the closed-loop poles of the actual closed-loop system and showing them on the root locus plot.

(B) Traditionally, there are two quick ways in dealing with the instability issue in (A). The first one is the so-called **gain stabilization**, i.e., reducing the loop gain at the high frequency large enough that the spill-over effect is significantly reduced for stability. This could be done by adding a lag compensation without using the low-frequency amplification factor (i.e., a lag compensator of $\frac{1}{n}\frac{s+n\alpha}{s+\alpha}$) or limiting the

closed-loop bandwidth as in the Method 2 in the motion control of hydraulic cylinder introduced in the lecture. For example, in the design of lead compensator without considering the flexible mode effect, instead of placing the dominant CL poles at -8±8j, let us place the three CL poles closer to the imaginary axis and the origin at -4±4j and -4. Verify that this will lead to the following lead compensator:

$$G_{Clead}(s) = \frac{0.0185(s+2.83)}{s+10.15} \tag{3}$$

Using MATLAB, obtain the closed-loop poles of the actual closed-loop system with this lead compensator and show them on the root locus plot. Verify the actual CL system is indeed stable and obtain the unit step response of the actual closed-loop system as well.

(C) The second one is the so-called **phase stabilization**, i.e., adding a pair of zeros near the resonance so as to shift the departure angles from the resonant poles pointing to LHP to cause the closed-loop poles in the LHP. This could be done by adding the following notch compensation:

$$G_{Cnotch}(s) = \frac{s^2 + 2\zeta_f \omega_f s + \omega_f^2}{\left(s+\omega_f\right)^2} \tag{4}$$

where ζ_f and ω_f are the damping ratio and the natural frequency of the notch filter. To see how this method works, let us first choose $\zeta_f = \zeta_p$ and $\omega_f = \omega_p$ so that the zeros of the notch compensation exactly cancel the lightly damped poles of the flexible modes. The lead compensator in (A) with the addition of this notch filter leads to the following controller:

$$G_C(s) = \frac{0.274(s+7.67)}{s+54.15} \cdot \frac{s^2+3.2s+80^2}{(s+80)^2} \tag{5}$$

Using MATLAB, obtain the closed-loop poles of the actual closed-loop system with this controller and show them on the root locus plot. Verify the actual CL system is indeed stable and obtain the unit step response of the actual closed-loop system as well.

(D) The phase stabilization in (C) via exact cancellation is quite sensitive to the knowledge of the resonance frequency and damping ratio of the flexible mode. In reality, these parameters especially the damping ratio ζ_p are hard to know exactly. To see the effect, let us assume that the actual resonance frequency and damping ratio are ω_p=84 and ζ_p=0.002 respectively. Using MATLAB, obtain the closed-loop poles of this actual closed-loop system with the controller in (5) and show them on the root locus plot. Verify that the actual CL system now becomes unstable.

(E) One way to make the notch compensation less insensitive to the knowledge of the resonance frequency and damping ratio of the flexible mode is to choose the notch frequency ω_f ***above*** the resonance frequency ω_p so that the departure angle from the lightly damped open-loop poles of the flexible mode points toward LHP. To see this, let us choose the same ζ_f as in (C) but with a larger ω_f of ω_f =85 in the design of the controller:

$$G_C(s) = \frac{0.274(s+7.67)}{s+54.15} \cdot \frac{s^2+2\times 0.02\times 85 s+85^2}{(s+85)^2} \tag{6}$$

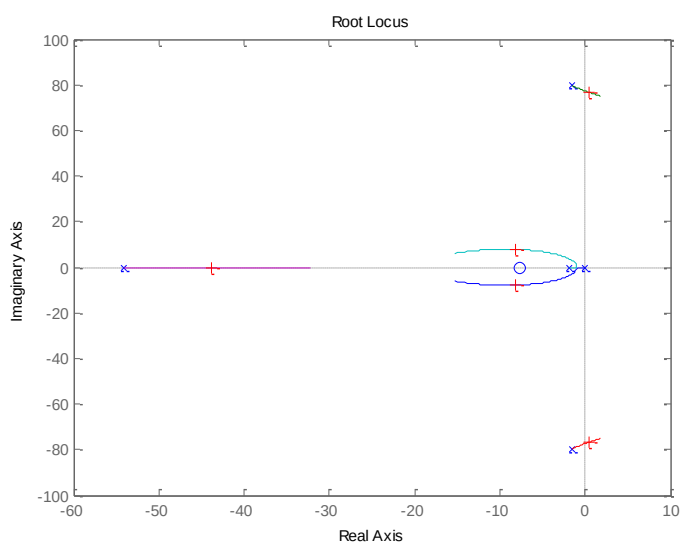
Using MATLAB, obtain the closed-loop poles of the actual closed-loop system with this controller and show them on the root locus plot for (i) ω_p=80 and ζ_p=0.02 as in (C); and (ii) ω_p=84 and ζ_p=0.002 as in (D)

respectively. Verify the actual CL system is indeed stable for both cases. Obtain the unit step response of the actual closed-loop system as well.

Solutions:

(A). With the given lead compensator, using the following MATLAB file, one can obtain the root locus plot of the design as shown in Fig.1, in which the pair of the CL poles due to the lightly damped flexible modes are in the RHP leading to instability.

```
clf; clear
J=0.00041; C=0.00076; omegap=80; zetap=0.02; Kp=1/C; taup=J/C; numP0=Kp; denP0=[taup,1,0];
numP=numP0*omegap^2; denP=conv(denP0,[1,2*zetap*omegap,omegap^2]);%plant
w1=-8+8*j; w2=-8-8*j; w3=-40; %desired CL pole locations
denCLd=poly([w1,w2,w3]); %desired CLCE
alpha1=denCLd(2)-1/taup;
Kc=(denCLd(3)-alpha1/taup)*taup/Kp;
beta1=denCLd(4)*taup/Kp/Kc;
numC0=Kc*[1,beta1]; denC0=[1,alpha1]; %The lead controller C(s)
[numL,denL]=series(numC0,denC0,numP,denP); %Form L(s)=CP, open-loop TF
[numT,denT]=cloop(numL,denL,-1); %Find the closed-loop TF T(s)
K=0:0.01:1.5;
rlocus(numL,denL,K);
[k,poles]=rlocfind(numL,denL)
```



k =

1.0090

poles =

0.4366 +76.7464i

0.4366 -76.7464i

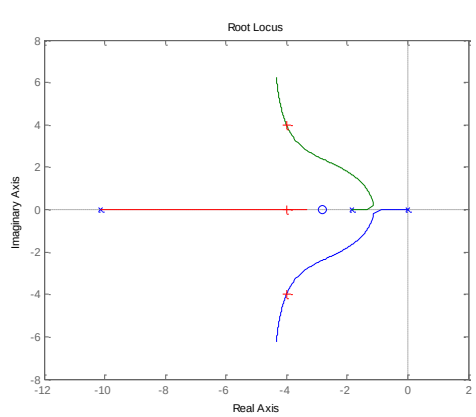
-43.7704

-8.1514 + 7.8610i

-8.1514 - 7.8610i

Fig.1

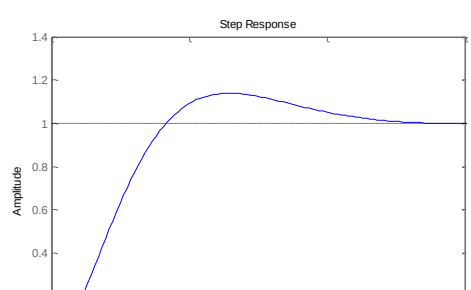
(B) When the flexible mode effect is neglected, as in (A) of Problem 1, the given lead compensator leads to the following unit step response and root-locus:



k =

0.9986

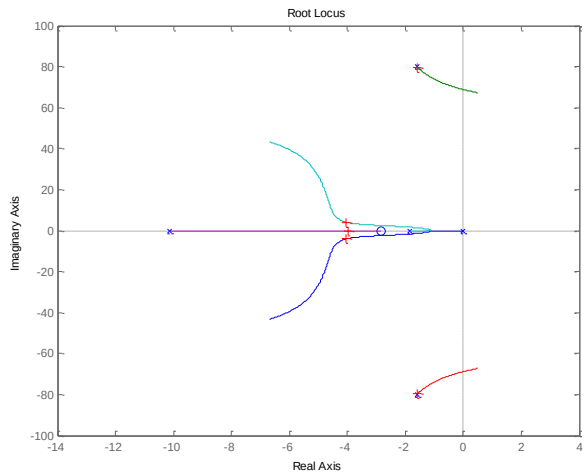
Fig.2



poles =
-3.9976 + 3.9919i
-3.9976 - 3.9919i
-4.0047

Fig.3

which indeed places the three CL poles at -4±4j and -4. When the flexible mode effect is considered as in (A), the root locus plot of the actual CL system with this lead compensator is shown in Fig.4, in which the pair of the CL poles due to the lightly damped flexible modes are close to the open-loop poles coming from the flexible modes, which are in the HP and stable though lightly damped. Since the dominant CL poles at -3.97 and -4.04±4.06j are much slower when compared with the lightly damped CL poles at -1.58±79.7j coming from the flexible modes, the unit step response of the actual CL system shown in Fig.5 reveals that no noticeable oscillations at the resonant frequency are seen.



k =
1.0092

poles =
-1.5785 +79.6987i
-1.5785 -79.6987i
-4.0355 + 4.0585i
-4.0355 - 4.0585i
-3.9719

Fig.4

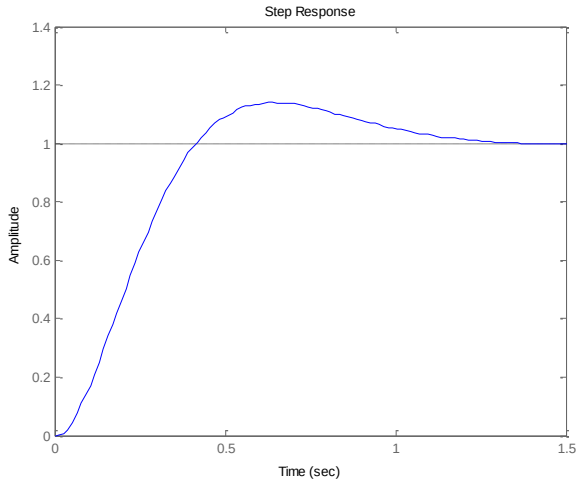
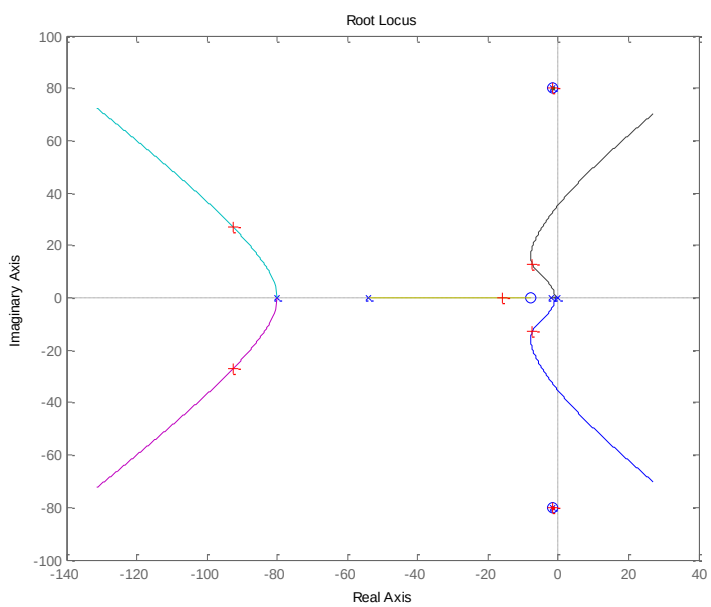


Fig.5

(C) Using the following M-file, the root locus plot of the actual CL system with the controller using notch filter is shown in Fig.6, in which one pair of the CL poles are exactly at the open-loop poles coming from the flexible modes due to pole/zero cancellation. It is seen that all 7 CL poles are stable and the dominant CL poles at -7.42±12.84j are near the targeted CL poles at -8±8j when synthesizing the lead compensator without considering the flexible mode effect. The unit step response of the actual CL system shown in Fig.7 reveals that no noticeable oscillations at the resonant frequency are seen as well.

```
clf; clear
J=0.00041;C=0.00076;omegap=80;zetap=0.02;Kp=1/C;taup=J/C;numP0=Kp;
denP0=[taup,1,0];
numP=numP0*omegap^2;denP=conv(denP0,[1,2*zetap*omegap,omegap^2]);%plant
w1=-8+8*j; w2=-8-8*j; w3=-40; %desired CL pole locations
denCLd=poly([w1,w2,w3]); %desired CLCE
alpha1=denCLd(2)-1/taup;
Kc=(denCLd(3)-alpha1/taup)*taup/Kp;
beta1=denCLd(4)*taup/Kp/Kc;
numC0=Kc*[1,beta1]; denC0=[1,alpha1]; %The lead controller C(s)
zetaf=zetap; omegaf=omegap;
numN=[1,2*zetaf*omegaf,omegaf^2];denN=[1,2*omegaf,omegaf^2]; %Notch filter
numC=conv(numC0,numN); denC=conv(denC0,denN);
[numL,denL]=series(numC,denC,numP,denP); %Form L(s)=C(s)P(s), open-loop TF
[numT,denT]=cloop(numL,denL,-1); %Find the closed-loop TF T(s)
step(numT,denT); %plot unit step response
pause
K=0:0.01:30;
rlocus(numL,denL,K);
[k,poles]=rlocfind(numL,denL)
```




```
k =
    0.9957

poles =
 -92.6051 +26.9344i
 -92.6051 -26.9344i
  -1.6000 +79.9840i
  -1.6000 -79.9840i
 -15.9502
  -7.4198 +12.8400i
  -7.4198 -12.8400i
```

Fig.6

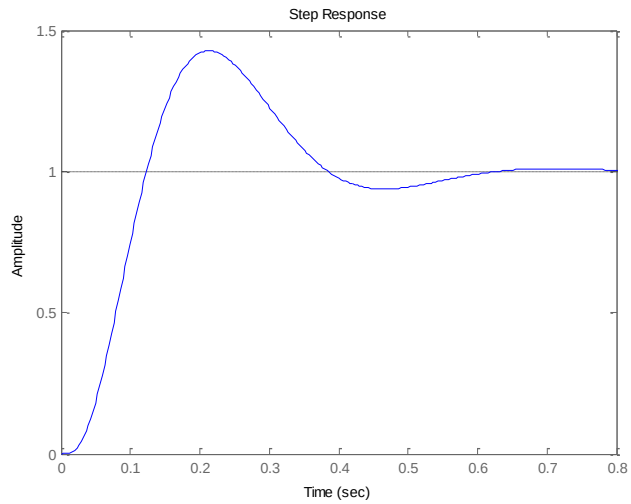
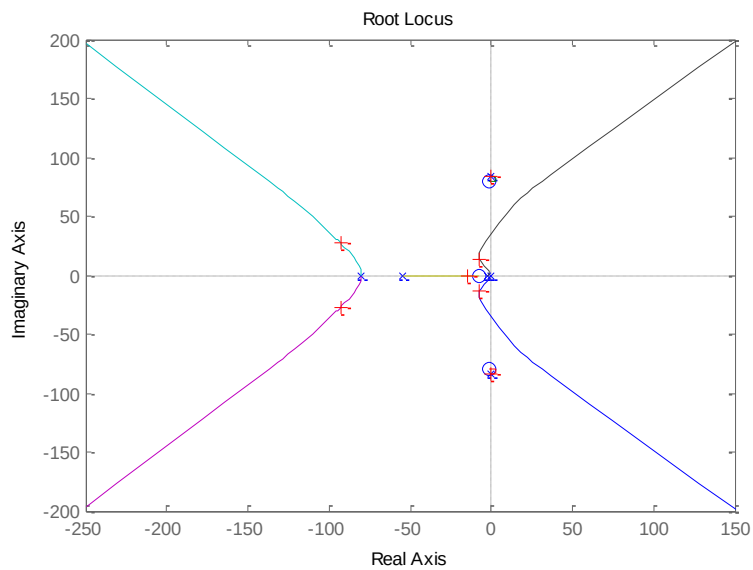


Fig.7

(D) As in (C), the root locus plot of the actual CL system is shown in Fig.8, in which the pair of the CL poles coming off the lightly damped open-loop poles of the flexible modes now becomes unstable at 0.0157±84.01j.



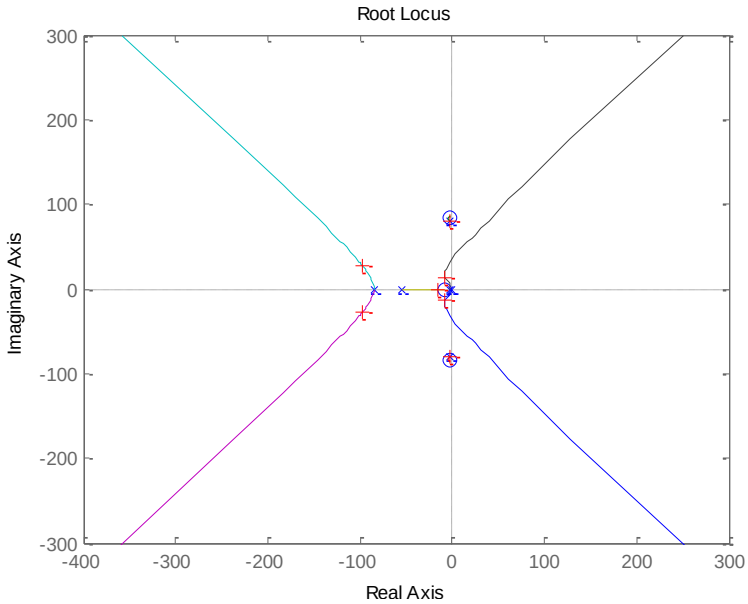
k =
1.0384

poles =
-92.9898 +27.7382i
-92.9898 -27.7382i
0.0157 +84.0118i
0.0157 -84.0118i
-7.6327 +13.4399i
-7.6327 -13.4399i
-15.1224

Fig.8

(E) **T**he root locus plot of the actual CL system for case (i) is shown in Fig.9 with the unit step response in Fig.10. **T**he root locus plot of the actual CL system for case (ii) is shown in Fig.11 with the unit step response in

Fig.12. In both cases all CL poles are stable and the unit step responses do not exhibit much oscillation at the resonant frequency.



k =

1.0377

poles =

-96.8790 +26.8919i

-96.8790 -26.8919i

-1.8021 +79.8723i

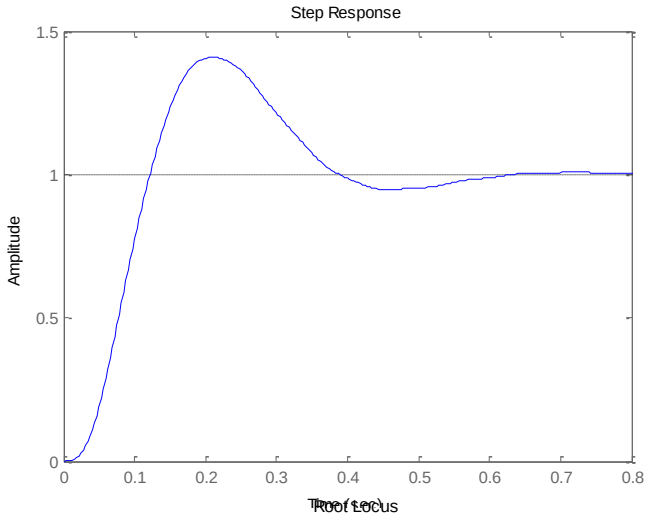
-1.8021 -79.8723i

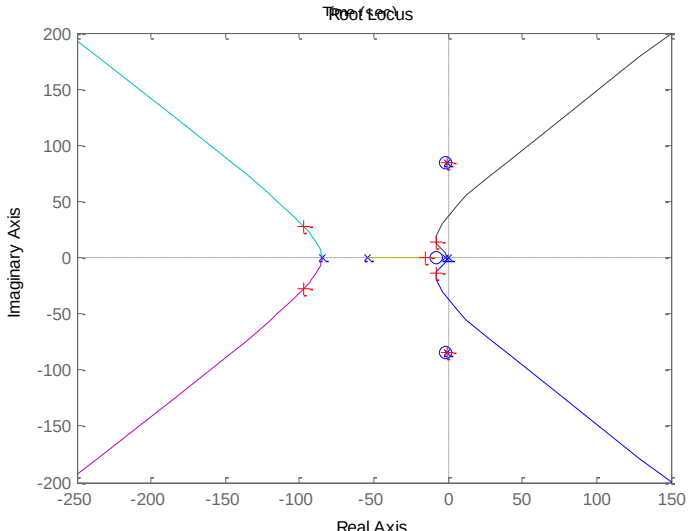
-15.7527

-8.0426 +13.3055i

-8.0426 -13.3055i

Fig.9





k =

1.0468

poles =

-97.0670 +27.4241i

-97.0670 -27.4241i

-0.1762 +83.9227i

-0.1762 -83.9227i
-15.6309
-8.1093 +13.3481i
-8.1093 -13.3481i

Fig.11

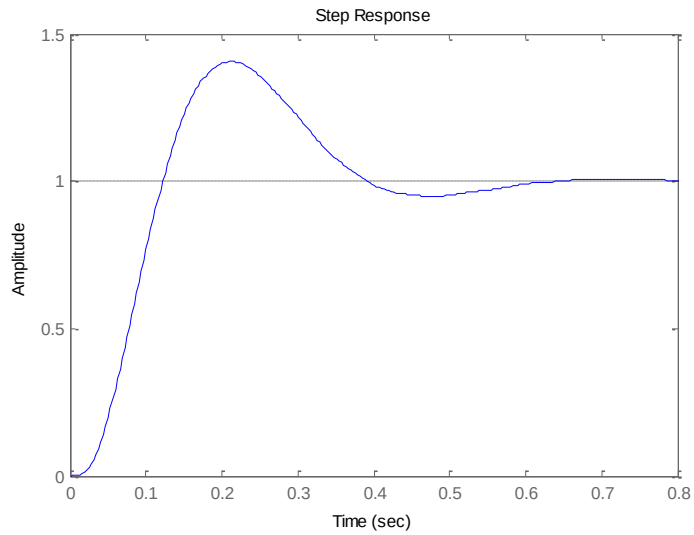


Fig.12

PROBLEM #3 (25%)

One problem with the phase stabilization via notch compensation in Problem 2 is that a pair of CL poles are always near the open-loop poles of the flexible mode which are very lightly damped. As seen in Problem 2, the effect of this pair of lightly damped CL poles on the unit step response is not much due to the near pole/zero cancellation effect – the zeros of the controller including those of the notch filter become the CL zeros for the transfer function from the reference input $R(s)$ to the output $\theta_2(s)$ as well. However, when the actual system has disturbances, their effect does show up as the zeros of the controller are not the CL zeros for the transfer function from the disturbance input $d(s)$ to the output $\theta_2(s)$ as shown below.

(F) Obtain the CLTF from d(s) to θ_2(s) for the controller in (E) of Problem 2 for ω_p=80 and ζ_p=0.01. Using MATLAB, obtain the unit step response and the unit disturbance response of the CL system as well and comment on the responses.

(G) To have a better disturbance response transient behavior, instead of the above quick fix using notch compensation, one could start the controller design process by directly dealing with the plant model that explicitly consider the flexible mode, i.e., the fourth order model in (1), as in the following. For example, even using a third-order controller with a proper transfer function given by

$$G_c(s) = \frac{\beta_3 s^3 + \beta_2 s^2 + \beta_1 s + \beta_0}{s^3 + \alpha_2 s^2 + \alpha_1 s + \alpha_0} \tag{7}$$

one would have seven controller parameters $\beta_3, \beta_2, \beta_1, \beta_0, \alpha_2, \alpha_1, \alpha_0$ to choose. Thus one should be able to arbitrarily place the seven CL poles when using this controller. Determine the controller

parameters that would place the seven CL poles at $p_{CLd1,2} = -8 \pm 8j$, $p_{CLd3} = -8$, $p_{CLd4,5} = -60 \pm 60j$, and $p_{CLd4,5} = -80 \pm 80j$ respectively for a plant with ω_p=80 and ζ_p=0.01.

(H) Using MATLAB, obtain the unit step response and the unit disturbance response of the CL system in (G) and compare the responses to those in (F).

Solutions:

(F). With the controller (6), the resulting CLTFs are:

$$\theta_2(s) = \frac{G_P G_c}{1+G_P G_c} R(s) + \frac{G_P}{1+G_P G_c} d(s) = \underbrace{\frac{N_P N_c}{D_P D_c + N_P N_c}}_{G_{CLR}(s)} R(s) + \underbrace{\frac{N_P D_c}{D_P D_c + N_P N_c}}_{G_{CLd}(s)} d(s) \quad (7)$$

It is thus clear that zeros of the controller, i.e., roots of the controller numerator N_c, are zeros of the CLTF $G_{CLR}(s)$ but not those of the CLTF $G_{CLd}(s)$. With the given values, these transfer functions are

$$G_{CLR}(s) = \frac{2.305e006s^3 + 2.551e007s^2 + 1.671e010s + 1.277e011}{0.5395s^7 + 122.8s^6 + 1.274e004s^5 + 1.022e006s^4 + 6.122e007s^3 + 1.482e009s^2 + 1.922e010s + 1.277e011}$$

$$G_{CLd}(s) = \frac{8.421e006s^3 + 1.888e009s^2 + 1.384e011s + 3.294e012}{0.5395s^7 + 122.8s^6 + 1.274e004s^5 + 1.022e006s^4 + 6.122e007s^3 + 1.482e009s^2 + 1.922e010s + 1.277e011}$$

Poles of CL0TFs are:

$$p_{CL1} = -16.766, \quad p_{CL2,3} = -7.866 \pm 12.605j, \quad p_{CL4,5} = -0.972 \pm 79.854j, \quad p_{CL4,5} = -96.579 \pm 26.483j$$

Zeros of $G_{CLR}(s)$ are:

$$z_{R1} = -7.669, \quad z_{R2,3} = -1.7 \pm 84.983j$$

Zeros of $G_{CLd}(s)$ are:

$$z_{d1} = -54.146, \quad z_{d2,3} = -85$$

It is seen that though there are a pair of lightly damped zeros $z_{R2,3} = -1.7 \pm 84.983j$ close to the lightly damped CL poles at $p_{CL4,5} = -0.972 \pm 79.854j$ in the CLTF $G_{CLR}(s)$, none of the zeros of the CLTF $G_{CLd}(s)$ are close to this pair of lightly damped CL poles in $G_{CLd}(s)$. As a result, no noticeable oscillations are observed in the unit step response shown in Fig.13 while oscillations around the resonance frequency are seen in the unit step disturbance response shown in Fig.14.

```
clf; clear
J=0.00041; C=0.00076; omegap=80; zetap=0.01; Kp=1/C; taup=J/C; numP0=Kp; denP0=[taup,1,0];
numP=numP0*omegap^2; denP=conv(denP0,[1,2*zetap*omegap,omegap^2]);%plant
w1=-8+8*j; w2=-8-8*j; w3=-40; %desired CL pole locations
denCLd=poly([w1,w2,w3]); %desired CLCE
alpha1=denCLd(2)-1/taup;
Kc=(denCLd(3)-alpha1/taup)*taup/Kp;
beta1=denCLd(4)*taup/Kp/Kc;
numC0=Kc*[1,beta1]; denC0=[1,alpha1]; %The lead controller C(s)
zetaf=0.02; omegaf=85;
```

```
numN=[1,2*zetaf*omegaf,omegaf^2];denN=[1,2*omegaf,omegaf^2]; %Notch filter
numC=conv(numC0,numN); denC=conv(denC0,denN);
plant=tf(numP,denP);
controller=tf(numC,denC);
[numL,denL]=series(numC,denC,numP,denP); OL=tf(numL,denL);
T=feedback(OL,1); %Find the closed-loop TF T(s)
step(numT,denT); %plot unit step response of lead controller with flexible mode
pause
K=0:0.01:10;
rlocus(numL,denL);
[k,poles]=rlocfind(numL,denL)
pause
Td=feedback(plant,controller);
step(Td);
pole(T); zero(T); pole(Td); zero(Td);
```

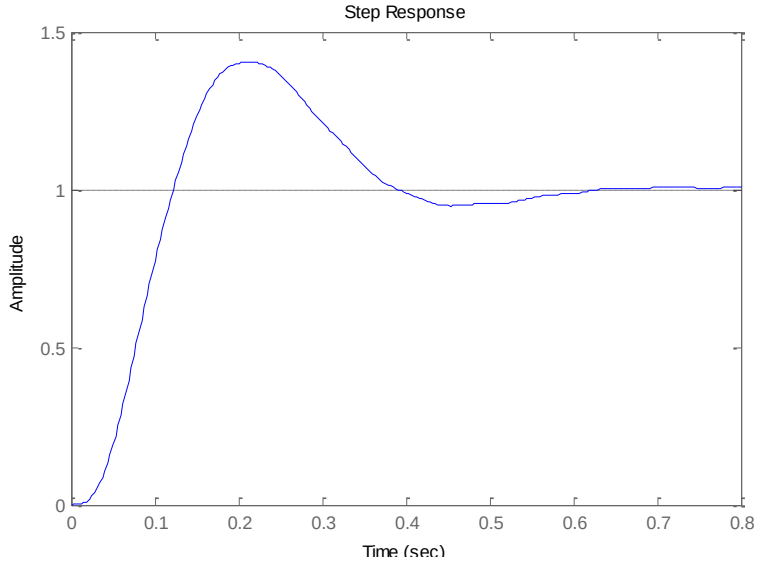


Fig.13 Unit step response

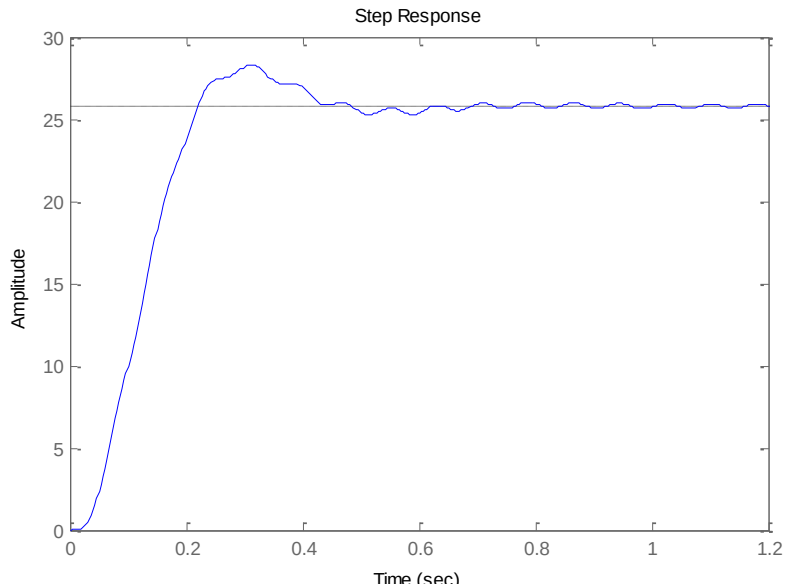


Fig.14 Unit disturbance input response

(G). To place all CL poles at $p_{CLd1,2} = -8 \pm 8j$, $p_{CLd3} = -8$, $p_{CLd4,5} = -60 \pm 60j$, and $p_{CLd4,5} = -80 \pm 80j$, the desired CLCE would be

$$\left((s+8)^2+8^2\right)(s+8)\left((s+60)^2+60^2\right)\left((s+80)^2+80^2\right) = s^7 + 396s^6 + 75208s^5 + 8112640s^4 + 514432000s^3 + 1.7234\text{e+}010s^2 + 2.1103\text{e+}011s + 1.2845\text{e+}012 \quad (8)$$

Noting that plant TF can be written as

$$G_P(s) = \frac{b_{P0}}{s^4 + a_{P3}s^3 + a_{P2}s^2 + a_{P1}s}, \qquad b_{P0} = K_p\omega_p^2 / \tau_p \quad (9)$$

Thus, with the controller (7), the actual CLCE is

$$\begin{aligned} D_P D_c + N_P N_c &= \left(s^4 + a_{P3}s^3 + a_{P2}s^2 + a_{P1}s\right)\left(s^3 + \alpha_2 s^2 + s + \alpha_0\right) + b_{P0}\left(\beta_3 s^3 + \beta_2 s^2 + \beta_1 s + \beta_0\right) \\ &= s^7 + \left(\alpha_2 + a_{P3}\right)s^6 + \left(\alpha_1 + a_{P3}\alpha_2 + a_{P2}\right)s^5 + \left(\alpha_0 + a_{P3}\alpha_1 + a_{P2}\alpha_2 + a_{P1}\right)s^4 + \\ &\left(b_{P0}\beta_3 + a_{P3}\alpha_0 + a_{P2}\alpha_1 + a_{P1}\alpha_2\right)s^3 + \left(b_{P0}\beta_2 + a_{P2}\alpha_0 + a_{P1}\alpha_1\right)s^2 + \left(b_{P0}\beta_1 + a_{P1}\alpha_0\right)s + b_{P0}\beta_0 = 0 \end{aligned} \quad (10)$$

Matching (8) and (10), we can solve for the controller as

$$G_C(s) = \frac{-5.78s^3 - 510.3s^2 + 447.8s + 6046}{s^3 + 300.5s^2 + 38740s + 1631000} = \frac{-5.78(s - z_{c1})(s - z_{c2})(s - z_{c3})}{(s - p_{c1})(s - p_{c2})(s - p_{c3})}, \quad (11)$$

$$z_{c1} = -89,\ z_{c2} = 3.82,\ z_{c3} = -3.08$$

$$p_{c1,2} = -113 \pm 95.5j,\ \ p_{c3} = -74.53$$

With this controller, the 7 CLTF poles are at the locations specified and

Zeros of $G_{CLR}(s)$ are:

$$z_{R1} = z_{c1} = -89, \quad z_{R2} = z_{c2} = 3.82, \quad z_{R3} = z_{c3} = -3.08$$

Zeros of $G_{CLd}(s)$ are:

$$z_{d1,2} = p_{c1,2} = -113 \pm 95.5j, \quad z_{d3} = p_{c3} = -74.53$$

Since there are no lightly damped CL poles, as seen from Figs.15 and 16, there are no high-frequency oscillations in both the unit step response and the unit step disturbance response. However, the CLTF $G_{CLR}(s)$ does have one RHP zero at $z_{R2} = 3.82$ due to the zero z_{c2} of the controller, which causes very large undershoot in the unit step response. Such an undesirable response problem could be solved by the judicious choice of the desired CL poles so that the resulting pole-placement controller does not have the RHP real zeros. For example, by choosing the desired locations for the 7 CL poles at $p_{CLd1,2} = -8 \pm 8j$, $p_{CLd3} = -500$, $p_{CLd4,5} = -60 \pm 60j$, and $p_{CLd4,5} = -300 \pm 300j$ instead of the locations used in (8), the resulting pole-placement controller would be

$$G_C(s) = \frac{784.7s^3 - 6866s^2 + 6.629e005s + 5.314e006}{s^3 + 1233s^2 + 6.282e005s + 1.555e008} = \frac{784.7(s - z_{c1})(s - z_{c2})(s - z_{c3})}{(s - p_{c1})(s - p_{c2})(s - p_{c3})},$$
$$z_{c1,2} = 7.91 \pm 29.9j, \quad z_{c3} = -7.08$$
$$p_{c1,2} = -303.4 \pm 395.6j, \quad p_{c3} = -625.8 \tag{12}$$

With this controller, the unit step response shown in Fig.17 does not exhibit undershoot and the unit step disturbance response shown in Fig.18 does not have the high-frequency oscillation as well.

It is also seen from Fig.16 that the steady-state error caused by the unit step disturbance is larger than that of in Fig. 14, though no high-frequency oscillations. Such a problem can be solved by using a higher-order controller having an integrator. For example, with a 4th -order controller of the form

$$G_c(s) = \frac{\beta_4 s^4 + \beta_3 s^3 + \beta_2 s^2 + \beta_1 s + \beta_0}{s\left(s^3 + \alpha_3 s^2 + \alpha_2 s + \alpha_1\right)} \tag{13}$$

the resulting CL system will be of type 1 w.r.t. the disturbance input, which will have zero steady error for unit step disturbance input. In addition, the 8 controller design parameters $\beta_4, \beta_3, \beta_2, \beta_1, \beta_0, \alpha_3, \alpha_2, \alpha_1$ also enable us to place the resulting 8 CL poles at any locations as we wish. For example, by choosing the desired locations for the 8 CL poles at $p_{CLd1,2} = -8 \pm 8j$, $p_{CLd3,4} = -60 \pm 60j$, $p_{CLd5,6} = -200 \pm 200j$ and $p_{CLd7,8} = -200 \pm 200j$ instead of the locations used in (8), the resulting pole-placement controller would be

$$G_C(s) = \frac{963.7s^4 + 5.194e004s^3 + 4.264e006s^2 + 5.731e007s + 3.779e008}{s^4 + 932.5s^3 + 4.284e005s^2 + 1.076e008s}$$
$$= \frac{963.7(s - z_{c1})(s - z_{c2})(s - z_{c3})(s - z_{c4})}{s(s - p_{c1})(s - p_{c2})(s - p_{c3})}, \quad z_{c1,2} = -19.6 \pm 56j, \; z_{c3,4} = -7.4 \pm 7.1j \tag{12}$$
$$p_{c1,2} = -213 \pm 408.8j, \quad p_{c3} = -506$$

With this controller, the unit step response shown in Fig.19 does not exhibit undershoot and the unit step disturbance response shown in Fig.20 has zero SS error and very small transient error as well, with no high-frequency oscillation!

(H)

```
clf; clear
J=0.00041; C=0.00076; omegap=80; zetap=0.01; Kp=1/C; taup=J/C; numP0=Kp/taup; denP0=[1,1/taup,0];
```

```
numP=numP0*omegap^2; denP=conv(denP0,[1,2*zetap*omegap,omegap^2]);%plant
w1=-8+8*j;w2=-8-8*j; w3=-60+60*j;w4=-60-60*j; w5=-200+200*j;w6=-200-200*j;w7=w5; w8=w6;%desired CL pole locations
denCLd=poly([w1,w2,w3,w4,w5,w6,w7,w8]); %desired CLCE
alpha2=denCLd(2)-denP(2);
alpha1=denCLd(3)-denP(2)*alpha2-denP(3);
alpha0=denCLd(4)-denP(2)*alpha1-denP(3)*alpha2-denP(4);
beta4=(denCLd(5)-denP(2)*alpha0-denP(3)*alpha1-denP(4)*alpha2)/numP;
beta3=(denCLd(6)-denP(3)*alpha0-denP(4)*alpha1)/numP;
beta2=(denCLd(7)-denP(4)*alpha0)/numP;
beta1=denCLd(8)/numP;
beta0=denCLd(9)/numP;
numC=[beta4,beta3,beta2,beta1,beta0]; denC=[1,alpha2,alpha1,alpha0,0]; %The C(s)
plant=tf(numP,denP);
controller=tf(numC,denC);
[numL,denL]=series(numC,denC,numP,denP); OL=tf(numL,denL);
T=feedback(OL,1); %Find the closed-loop TF T(s)
step(T); %plot unit step response of lead controller with flexible mode
pause
K=0:0.01:10;
rlocus(numL,denL);
[k,poles]=rlocfind(numL,denL)
pause
Td=feedback(plant,controller);
step(Td);
pole(T); zero(T); pole(Td); zero(Td);
```

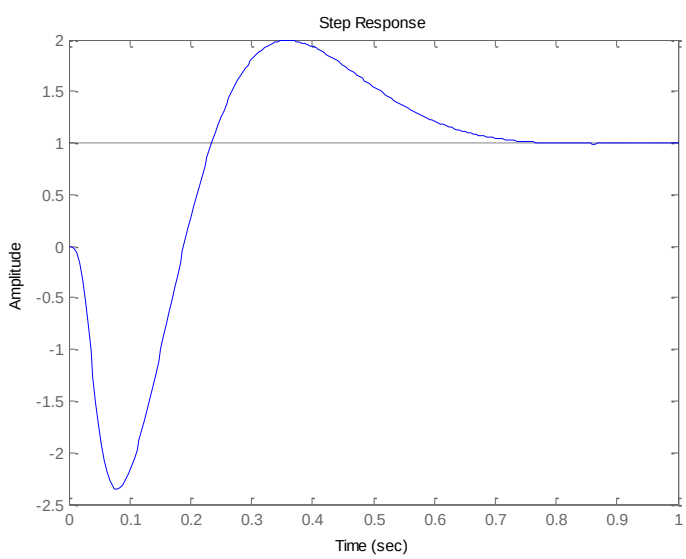


Fig.15 Unit step response

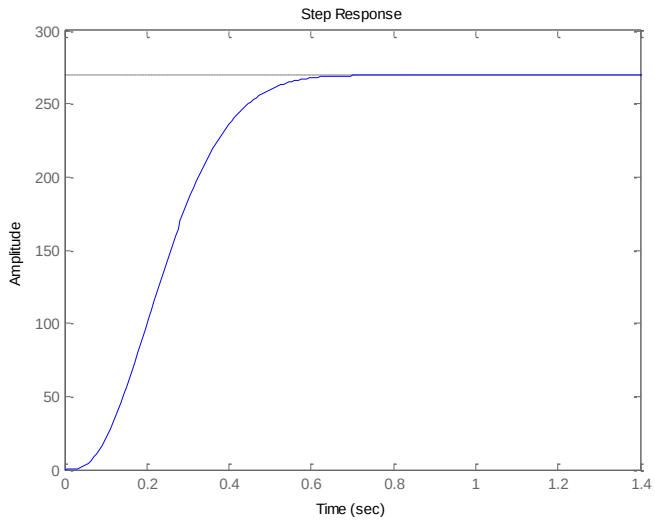


Fig.16 Unit step disturbance response

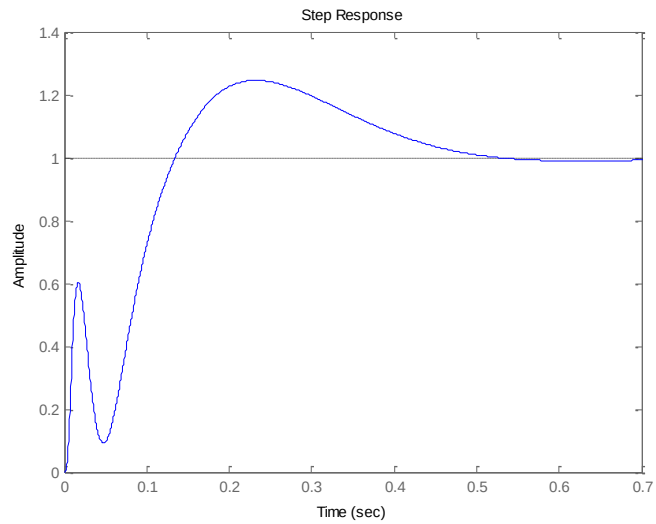


Fig.17 Unit step response

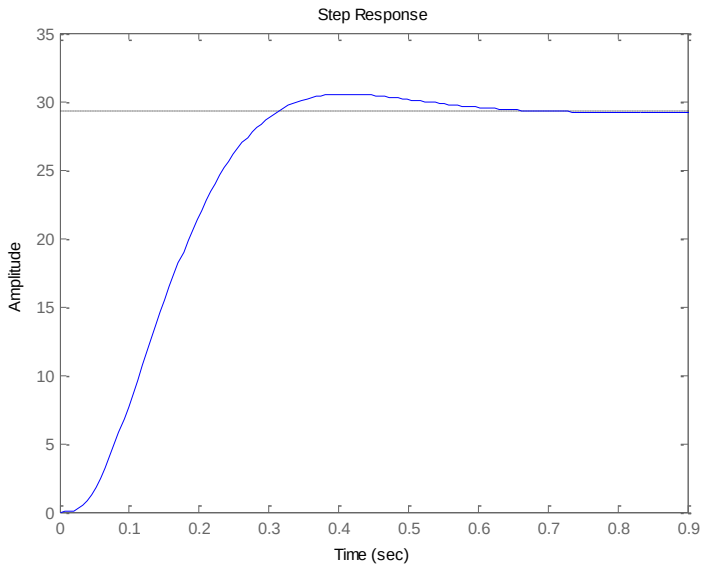


Fig.18 Unit step disturbance response

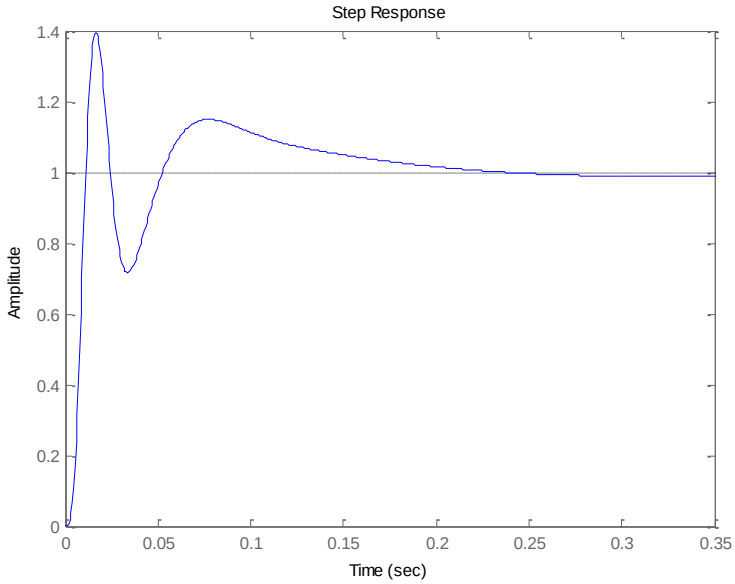


Fig.19 Unit step response

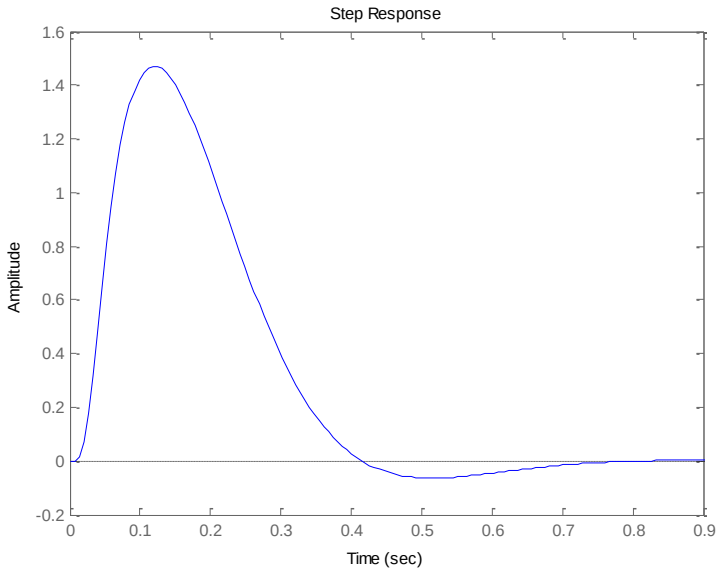


Fig.20 Unit step disturbance response